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Alternative data for restaurant bankruptcy prediction

Pascal Kündig^{1,2*}

Abstract

The restaurant industry has a high failure rate and predicting restaurant bankruptcy is an important task to mitigate economic losses. Restaurant consumer review data contains rich information and influences future restaurant demand. We investigate the value of publicly available data for predicting bankruptcies of individual restaurants. Our unique data set includes alternative consumer data from the two most frequently used online restaurant platforms in Switzerland and official business data from Swiss government websites. In evaluating the accuracy of predictive bankruptcy probabilities, we quantify the potential of the alternative data sources and we analyze the relevance of novel predictor variables. We find that alternative consumer data is relevant for restaurant bankruptcy prediction and the highest prediction accuracy is achieved when alternative data and traditional business data are combined.

Keywords Risk analysis, Public data, Web scraping

1 Introduction

Online platforms such as Tripadvisor and Google allow consumers to share their experiences and the resulting data can contain rich information about a restaurant's success. Such alternative consumer data covers various aspects of restaurants, such as the quality of the food and service, the value for money, or the hygiene (Farronato and Zervas, 2022). In addition to providing information about the current situation, this data is also crucial for the future success of restaurants. Often, consumers face ex-ante uncertainty about the quality of the restaurant. Reviews can resolve this information asymmetry and therefore guide future demand (Anderson and Magruder, 2012; Luca and Zervas, 2016). Consumer data also influences the restaurant's future business environment, e.g., a higher review rating can improve a restaurant's future loan terms and performance (Huang, 2024).

The restaurant industry has a high failure rate (Hensler et al., 1997; Gu, 2002), and predicting bankruptcies of individual restaurants is therefore an important task to mitigate economic losses. However, traditional bankruptcy models in general (Altman, 1968; Olsen et al., 1983; Zmijewski, 1984; Shumway, 2001; Bellovary et al., 2007) and for restaurants (Gu, 2002; Kim and Gu, 2006; Youn and Gu, 2010; Kim and Upneja, 2014; Becerra-Vicario et al., 2020) mainly use predictor variables based on financial ratios and basic characteristics of the business. Alternative consumer data contains complementary information to this traditional business data, may be more accessible and up-to-date, and has the potential to improve the prediction accuracy of bankruptcy models. We contribute to this open field of research by collecting publicly available consumer and business data for restaurants in Switzerland and by predicting bankruptcies based on the collected data. In evaluating the accuracy of predictive bankruptcy probabilities, we quantify the potential of the alternative data sources and we analyze the relevance of novel predictor variables through model interpretation. Our findings have direct implications for lenders, investors, and policymakers, as more accurate bankruptcy models lead to better decision-making.

*Correspondence:

Pascal Kündig
pascal.kuendig@hslu.ch

¹ Lucerne School of Business, Lucerne University of Applied Sciences and Arts, Rotkreuz, Switzerland

² Faculty of Business and Economics, University of Basel, Basel, Switzerland

In the general bankruptcy and credit risk literature of recent years, there are several proposals that use different alternative data to improve the prediction accuracy of bankruptcy models (Liang et al., 2016; Jo and Shin, 2016; Mai et al., 2019; Kou et al., 2021; Djeundje et al., 2021). In this context, our research is a specialization that focuses on consumer data as alternative data and bankruptcy prediction is conducted exclusively for restaurants. There is also work on using alternative consumer data to model restaurant closures and growth (Lian et al., 2017; Te et al., 2018; Tao and Zhou, 2020; Li et al., 2023; Zhang and Luo, 2023), but to the best of our knowledge, no alternative consumer data has been used to model restaurant bankruptcies. Note that restaurant bankruptcy is fundamentally different from restaurant closure, as bankruptcy is a clearly defined legal process that occurs when the business is no longer able to pay its debts. In contrast, restaurant closure is a decision taken by the owner for various reasons, such as poor performance, financial difficulties, or personal circumstances. Further, in the case of a restaurant chain, bankruptcy affects the whole company, whereas the closure of one restaurant does not necessarily have a direct impact on the other restaurants in the chain.

In this article, we focus on modeling bankruptcies of individual restaurants, and our data set includes a few restaurants that are part of a chain. We collect alternative consumer data from the two most frequently used online restaurant platforms in Switzerland, Tripadvisor and Google, as well as official business data from Swiss government websites. Combining these two publicly available data sources allows for the identification of restaurant bankruptcies and the construction of multiple predictor variables, with several variables relying on collected consumer reviews. In our bankruptcy prediction study, we find that it is possible to discriminate between restaurants that will go bankrupt and those that will not by only using alternative consumer data. Specifically, we obtain an AUC of 0.677 with a logistic regression model that includes only predictor variables based on data from the restaurant platforms. The prediction accuracy increases when alternative consumer data is combined with traditional business data, i.e., we obtain an AUC of 0.729 with a logistic regression model that includes all available predictor variables. With a non-linear random forest model, the prediction accuracy can be improved a little further; in particular, we obtain an AUC of 0.733. We find that the number of consumer reviews and the average review rating on Google over the last year are highly relevant predictor variables in our bankruptcy models. In addition, measuring the local competition by the difference in the average consumer rating between the neighborhood and the restaurant is effective and we also find interaction

effects among predictor variables based on official business data and predictor variables based on alternative consumer data.

The remainder of this article is organized as follows. Section 2 discusses the related literature and Sect. 3 describes the data collection procedure, the selection of restaurants, the identification of bankruptcies, and the predictor variables. In Sect. 4, we perform restaurant bankruptcy prediction, evaluate the prediction accuracy and model quality, and analyze the relevance of novel predictor variables. Section 5 comments on the limitations of our research and Sect. 6 concludes.

2 Literature

Altman (1968) pioneered the field of bankruptcy prediction by identifying five key financial ratios and using multiple discriminant analysis to predict bankruptcy among manufacturing firms. Many other models have been introduced for predicting bankruptcy, such as the logistic regression model (Ohlson, 1980; Altman et al., 2008) and later non-linear machine learning models (Zieba et al., 2016; Barboza et al., 2017). However, the predictor variables used in these bankruptcy models are mainly based on financial ratios and basic characteristics of the business (Bellovary et al., 2007). While there exists an extensive literature on developing bankruptcy models, prediction accuracy can also be increased by using new data sources for predictor variables. The role of non-financial data for credit risk was first investigated by Grunert et al. (2005) and later by Altman et al. (2008). Both studies find that model accuracy improves when non-financial data is used in addition to financial ratios. In the general bankruptcy and credit risk literature, there are now several proposals that focus on different types of alternative data, i.e., Liang et al. (2016) include corporate governance indicators, Jo and Shin (2016) create sentiment variables from public news articles, Mai et al. (2019) extract information from financial text data, and Kou et al. (2021) use payment and transaction data.

Like general bankruptcy models, restaurant bankruptcy models traditionally include financial ratios as predictor variables (Gu, 2002; Kim and Gu, 2006; Youn and Gu, 2010; Kim and Upneja, 2014; Becerra-Vicario et al., 2020). However, Parsa et al. (2005) emphasize that the success or failure of a restaurant depends on a variety of factors, and these factors are not fully reflected in financial ratios. For restaurant closure and growth, several studies have helped to reduce this discrepancy by using alternative consumer review data. In particular, Lian et al. (2017) model restaurant closure using publicly available restaurant data from China and the predictor variables are based on collected location characteristics, user mobility, review ratings, and review texts. Te et al.

(2018) model the growth of restaurants in Switzerland, combining publicly available consumer data with non-public data from an insurance company. Other studies that use consumer review data for restaurant closure extract consumer sentiment from review texts (Tao and Zhou, 2020; Li et al., 2023) and information from photos posted by consumers (Zhang and Luo, 2023).

The claim that alternative consumer data contains relevant information for restaurant bankruptcy prediction is also supported by causal economic research. Specifically, Anderson and Magruder (2012) provide evidence that higher Yelp ratings cause restaurants in San Francisco to sell out more often during prime time, and Luca (2016) identifies a positive causal effect of Yelp ratings on restaurant revenues. Farronato and Zervas (2022) find causal evidence that negative hygiene signals in consumer reviews reduce restaurant consumer demand. Further, Huang (2024) studies loan outcomes for small businesses, including restaurants, and finds that a higher average review rating on Yelp significantly reduces the likelihood of a loan default, reduces the collateral required, and lowers the interest rate charged on the loan. Collectively, these studies demonstrate the influence of consumer review data on a restaurant's future demand and business environment - critical factors that can determine success or failure.

3 Data

3.1 Data collection

Using web scraping techniques, we first collect a comprehensive set of 21'887 restaurants in Switzerland from Tripadvisor ([tripadvisor.ch](https://www.tripadvisor.ch)). For each restaurant, we collect the restaurant name, the address, the website address, the phone number, the price category, the total number of consumer reviews and the average consumer review rating over all previous years, and additional ratings concerning the quality of the kitchen, the quality of the service, and the value for money. For our set of restaurants, we also collect all available 1'156'600 consumer reviews on Tripadvisor with the review date, rating, and text. We then successfully re-identify 20'861 of these restaurants on Google ([google.ch](https://www.google.ch)) using the restaurant name, address, phone number, and website address as matching criteria. On Google, we collect for each of the re-identified restaurants the price category, general characteristics of the restaurant in text form, the total number of consumer reviews and the average consumer review rating over all previous years, and the latitude and longitude coordinates of the restaurant. For 19'316 restaurants, we also collect all 4'227'198 consumer reviews available on Google with the review date, rating, and text.

After collecting consumer data on Tripadvisor and Google, the Swiss Federal Statistical Office (FSO) adds

the enterprise identification number (UID) to our set of restaurants, using the restaurant's name and address as identification criteria. Every company operating in Switzerland is assigned a UID, and a company can have several local businesses. To determine the correct UID of a restaurant, restaurants must be identified at the local business level using data from the Business and Enterprise Register. From our set of restaurants, 8'656 restaurants can be uniquely identified as a local business and the UID is successfully added for these restaurants. We further collect official business data from the public UID-register for the restaurants with added UID. In particular, we store the legal form of the restaurant and, in the case that the business is liable to value-added tax (VAT), we also store the start date of the VAT liability and, if available, the end date of the VAT liability. The final data source we consider is the Swiss Official Gazette of Commerce (SOGC). All bankruptcy proceedings in Switzerland are published in the SOGC, and we collect all available bankruptcy publications for businesses. In Sect. 3.2, we describe how bankruptcies of restaurants are identified.

Figure 1 shows the described data collection procedure for our restaurant bankruptcy data set. Data collection on Tripadvisor and Google, excluding consumer review collection on Google, was conducted in February 2022. Consumer reviews on Google were collected in March 2022 and the FSO added the UID in August 2022. In June 2024, data from the UID-register and bankruptcy publications of the SOGC were collected.

3.2 Bankruptcy identification and restaurant selection

We impose several restrictions on the restaurants in the final data set to successfully identify restaurant bankruptcies. First, restaurants must be uniquely identified as local businesses with an assigned UID. Bankruptcy publications of the SOGC always include the UID of the bankrupt business if it is either a public limited company (PLC) or a limited liability company (LLC). For other legal forms, such as the sole proprietorship, the business owner is subject to bankruptcy proceedings, not the business itself, and the UID is either unpublished or contained in an unstructured text. Therefore, it is only possible to determine with certainty whether a restaurant has gone bankrupt if it is either a PLC or an LLC, and we limit our set to these types of restaurants. The limitation to PLCs and LLCs is an important constraint on the results of our research. Traditional business data is potentially more readily available for PLC and LLC restaurants than for restaurants with other legal forms, such as the wide spread sole proprietorship. Thus, it may well be that alternative consumer data is particularly valuable for predicting bankruptcies of restaurants that are not included in our final data set. Also, bankruptcies may

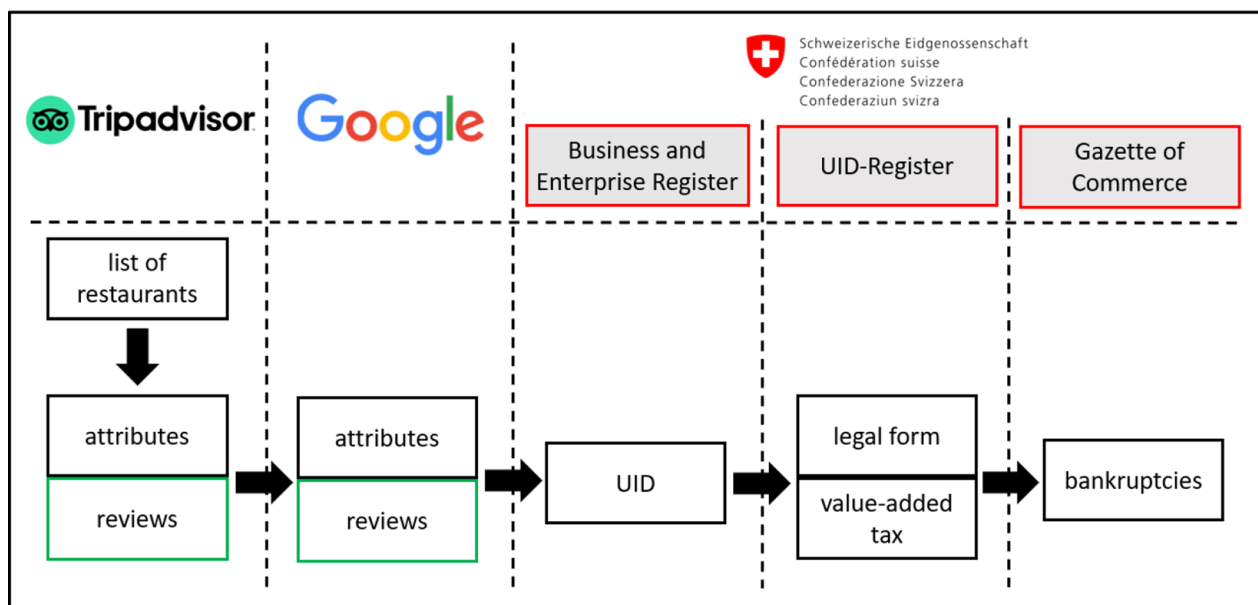


Fig. 1 Data collection procedure

be more common in this not-analyzed set of restaurants and the economic gains from more accurate bankruptcy models would be even greater for these restaurants.

From the collected list of bankruptcy publications, we select those that contain a UID and determine a bankruptcy date for each publication. If available, the bankruptcy date is equal to the date when bankruptcy proceedings opened; otherwise, it is equal to the bankruptcy publication date. When there are multiple bankruptcy publications for a UID, we select the one with the earliest bankruptcy date. A bankruptcy of a restaurant is identified if the restaurant's UID is included in our list of bankruptcies. We also require that the restaurants are liable to VAT at the beginning of the period for which we conduct bankruptcy prediction. A business in Switzerland is liable to VAT if its annual turnover reaches the threshold of 100'000 CHF.¹ Since the VAT liability only exists during the period of business activity, this restriction ensures that the restaurant businesses are active at the beginning of the prediction period. Discrepancies between a restaurant's bankruptcy date and the end date of its VAT liability can occur. If the bankruptcy date is earlier than the VAT end date, we assign the former to the latter. We also do this, if the bankruptcy date is no

more than one year after the VAT end date. If a restaurant goes bankrupt, but there is no stored VAT end date, we set the bankruptcy date as the VAT end date. To construct the predictor variables described in Sect. 3.3, we also require that the restaurants have been successfully re-identified on Google and that their consumer reviews on Google have been collected.

Since data collection on Tripadvisor and Google was conducted in February 2022, we predict bankruptcies for the two-year period from March 2022 to March 2024. Our data set includes all restaurants that meet the restrictions described in this Sect. and are liable to VAT at the beginning of March 2022. In total, our data set contains 4'275 restaurants, of which 204 restaurants went bankrupt in the two-year period from March 2022 to March 2024. Figure 2 shows the number of monthly defaults for this period, and Fig. 3 shows the locations of the restaurants in Switzerland, with red dots indicating restaurants that went bankrupt during the prediction period. Many restaurants are located in the urban areas of Zurich, Lucerne, Basel, Bern, Geneva, and Lugano, and only a few restaurants are located in the mountain regions.

3.3 Predictor variables

We construct predictor variables based on the alternative consumer data collected from Tripadvisor and Google, and the official business data collected from the UID-register. Several predictor variables rely on consumer reviews from the restaurant platforms. We transform the Google review ratings to obtain similar

¹ VAT liability starts with the business activity if the turnover threshold is expected to be reached in the first twelve months. Businesses that were not previously liable to VAT will become liable after the end of the financial year in which the turnover threshold is reached.

VAT liability ends with the termination of business activity or with the completion of the liquidation process. If the annual turnover does not reach the threshold and is not expected to be reached in the following tax period, the business can exempt itself from VAT liability for the following tax periods.

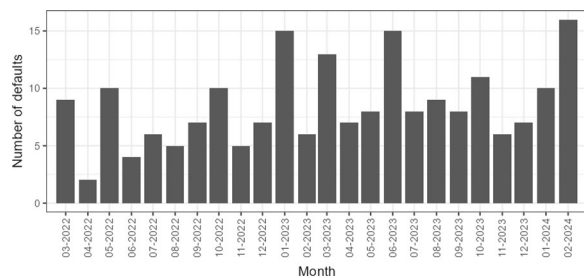


Fig. 2 Number of monthly defaults

rating scales for the Tripadvisor and Google reviews. To avoid a potential forward-looking bias, we only consider reviews written in or before January 2022. A German review text is available for almost every second review. We predict sentiment scores between 0 (= negative) and 100 (= positive) for these reviews using a language model based on the Google BERT architecture and specifically trained on German language samples (Guhr et al., 2020). Table 6 in Appendix A.2 shows the sentiment scores for a positive, a neutral, and a negative review text. For each restaurant, we count the number of reviews and calculate the average review rating and sentiment score on Tripadvisor and Google, respectively, for the year, the two years, and the five years prior to the beginning of the prediction period in March 2022. We also construct predictor variables for the total number of reviews and the average review

rating over all previous years, but these variables are based on the values displayed directly on the restaurant websites on Tripadvisor and Google, not on the collected consumer reviews.

Google provides general characteristics of the restaurants in text form, and we create dummy variables for the fifteen most common ones in our restaurant set to classify a restaurant type. These variables indicate the kind of food that is served and whether the restaurant is a café, bar, bistro, or take-away. We create predictor variables from the Tripadvisor ratings for the quality of the kitchen, the quality of the service, and the value for money. Further, we include the restaurant price categories as predictor variables from both Tripadvisor and Google. Based on the collected business data from the UID-register, we create a variable that indicates whether the restaurant is a PLC or a LLC, and we calculate the number of days for which the restaurant is already liable to VAT at the beginning of March 2022. To identify restaurants that are part of a chain, we create a variable that indicates whether a restaurant's UID occurs more than once in our data set.

We transform the restaurant coordinates collected on Google to the Swiss coordinate system LV95 and determine for each restaurant the twenty closest restaurants in terms of Euclidean distance. As potential neighbors, we consider the large set of 20'861 restaurants that could be re-identified on Google. Predictor variables capturing

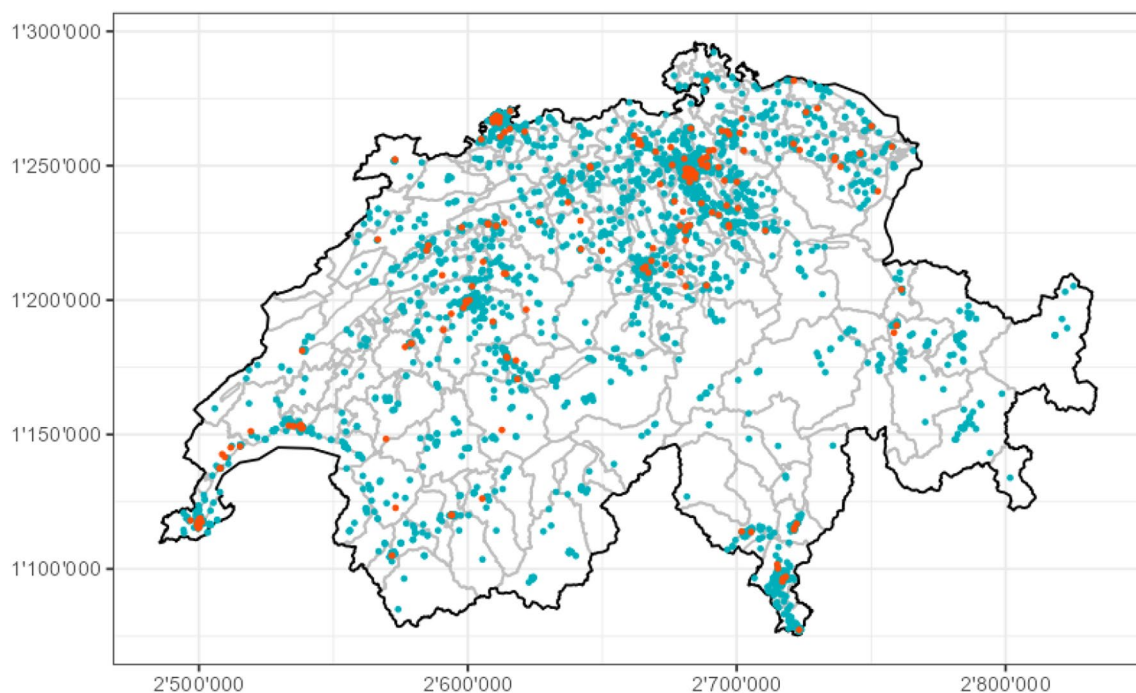


Fig. 3 Locations of the restaurants in Switzerland. Red dots indicate restaurants that went bankrupt

the local competition have been identified as highly relevant when predicting restaurant closure (Tao and Zhou, 2020; Li et al., 2023), and to measure the local competition, we compute averages over the neighbor set of each restaurant. In particular, we calculate the average spatial distance to the neighbors. Over the neighbor set, we also calculate the average number of reviews and the average review rating over all previous years on Tripadvisor and Google, respectively. From these averages, we then subtract the restaurant's number of reviews and average review rating to get differences between the restaurant and its neighborhood. Further, to capture the similarity of neighboring restaurants, we count the number of restaurants in the neighbor set that belong to the same classified restaurant type.

We also include demographic variables that are made publicly available by the FSO. Specifically, at the municipal level, we include the number of residents and the number of employees in the first, second, and third sectors. At the cantonal level, we include the tax on profits in percent, and at the greater region level, which covers several cantons, we include the median gross monthly wage. Finally, our set of predictor variables is extended by some interaction variables between predictor variables based on official business data and predictor variables based on alternative consumer data. A description of the predictor variables is provided in Table 7 in Appendix A.3. Missing values for numeric predictor variables are imputed using the mean and restaurants with missing information on the price category are assigned the category zero. Summary statistics for the numeric and categorical variables are provided in Tables 8 and 9, respectively, in Appendix A.4. We additionally include an intercept term in the predictor variables.

4 Prediction of individual bankruptcy probabilities

We predict bankruptcy probabilities of individual restaurants for the two-year period from March 2022 to March 2024. Bankruptcy prediction is performed on our unique data set using different models and sets of predictor variables. The accuracy of predictive bankruptcy probabilities is evaluated using a stratified 5-fold cross-validation procedure, i.e., we make out-of-sample predictions. For each model and its pooled predictive bankruptcy probabilities of all cross-validation loops, we evaluate the following prediction accuracy measures: the area under the receiver operating characteristic curve (AUC), the H-measure, the mean log-loss, the Brier score, and the expected calibration error (ECE). The AUC measures predictive discrimination ability,

and it can be interpreted as the probability that the predictive probability for a randomly drawn restaurant with a bankruptcy event is higher than the predictive probability for a randomly drawn restaurant without a bankruptcy event. The ECE assesses calibration. A predictive probability p is calibrated if the corresponding event (= bankruptcy) occurs in $100 \times p$ percent. We determine the boundaries of the bins for the ECE by using 20 equally-spaced quantiles of the empirical distribution of all predictive probabilities of all models from Sect. 4.1, respectively, from Sect. 4.2, pooled together. The H-measure, the log-loss, and the Brier score measure overall predictive accuracy. See, e.g., Dimitriadis et al. (2023) for more information on these prediction accuracy measures.

4.1 Model comparison using all predictor variables

We perform bankruptcy prediction with a logistic regression model, a lasso model, a linear spatial Gaussian process (GP) model, a tree-boosting model, and a random forest model. Each model includes all available predictor variables from Table 7 in Appendix A.3. We use the `glmnet` library version 4.1-8 for the lasso model and the `ranger` library version 0.17.0 for the random forest model. For the linear spatial GP model and the tree-boosting model, we use the `GPBoost` library version 1.5.3. Note that the tree-boosting implementation of the `GPBoost` library is based on the `LightGBM` library. Parameter tuning for the lasso, tree-boosting, and random forest models is repeated on the training data set of each cross-validation fold. For the lasso and the tree-boosting models, the tuning parameters are selected using an inner five-fold cross-validation procedure on the training data set and the candidate tuning parameters for the tree-boosting model are shown in Table 5 in Appendix A.1. For the random forest model, the tuning parameters are selected using the procedure from the `tuneRanger` library version 0.7 on the training data set and we tune the fraction of predictor variables as splitting candidates, the sampling fraction of the observations, and the minimum node size and set the number of trees to 500. In general, we use the AUC as the tuning criterion.

Table 1 reports the AUC, H-measure, mean log-loss, Brier score, and ECE for each model. Overall, the differences in the prediction accuracy measures between the models are small, i.e., the models achieve a similar prediction accuracy. Compared to the simplest model, the logistic regression model, the additional regularization in the lasso model leads to more accurate predictive bankruptcy probabilities in terms of H-measure, mean log-loss, Brier score, and ECE, but not in terms of AUC. For the linear spatial GP model, only a variance parameter of

Table 1 AUC, H-measure, mean log-loss, Brier score, and ECE

	AUC	H-measure	Mean log-loss	Brier score	ECE
Logistic regression	0.7291	0.1686	0.1902	4.470×10^{-2}	1.486×10^{-2}
Lasso	0.7269	0.1735	0.1760	4.374×10^{-2}	1.024×10^{-2}
Linear spatial GP	0.7298	0.1669	0.1851	4.468×10^{-2}	1.521×10^{-2}
Tree-boosting	0.7300	0.1708	0.1772	4.401×10^{-2}	1.206×10^{-2}
Random forest	0.7329	0.1631	0.1775	4.402×10^{-2}	1.349×10^{-2}

The best value for each prediction accuracy measure is highlighted in bold

$\sigma^2 = 1.862 \times 10^{-3}$ and a range parameter of $\rho = 127/838$ are estimated on the full data set. Estimating a small variance parameter indicates small spatial residual correlation in the data, and the linear spatial GP model therefore does not outperform the logistic regression model. Further, the prediction accuracy of the non-linear tree-boosting and random forest models is slightly higher than that of the logistic regression model, except for the smaller H-measure of the random forest model.

As the prediction accuracy of the logistic regression model is competitive, and as the logistic regression model allows for straightforward inference of the model coefficients, we use it in the following to analyze the predictor variables.

4.2 Analysis of predictor variables using logistic regression

We perform bankruptcy prediction based on nine different sets of predictor variables using logistic regression. Model quality and coefficients are evaluated when model estimation is conducted on the full data set, and we assess model quality using the Akaike Information Criterion (AIC). The AIC rewards goodness of fit, measured by the likelihood, and penalizes model complexity, measured by the number of predictor variables. Table 2 reports the AIC, AUC, H-measure, mean log-loss, Brier score, and ECE for all nine logistic regression models. The full

sets of estimated model coefficients, including standard errors, for all nine models are reported in Tables 10 and 11 in Appendix A.5. For a better understanding, we show in Tables 3 and 4 the most important model coefficients for the logistic regression models (1) to (3) and (5) to (7), respectively, where the coefficients for the coordinates, the restaurant types, and the price categories are only indicated. Significant model coefficients at the five percent level are always marked with an asterisk.

In the following, we use model (1) as our reference model, which includes the following predictor variables: the indicator for public limited companies (PLC), the number of days for which the restaurant business is already liable to VAT (VAT_days), the indicator for restaurant chains (chains), the spatial location of the restaurant (longitude, latitude), indicators for the type of food that is served (swiss_kitchen, italian_kitchen, thai_kitchen, chinese_kitchen, indian_kitchen, asian_kitchen, japanese_kitchen, pizza, burger, sushi, steak), indicators whether the restaurant is a café, bar, bistro, or take-away (cafe, bar, bistro, take_away), the price categories on Google and Tripadvisor (Gprice, Tprice), the rating for the quality of the kitchen (Trating_kitchen), the quality of the service (Trating_service), the value for money (Trating_value), the average sentiment score on Google and Tripadvisor over the last year (Gsentiment_lastyear,

Table 2 AIC, AUC, H-measure, mean log-loss, Brier score, and ECE for the nine logistic regression models

	AIC	AUC	H-measure	Mean log-loss	Brier score	ECE
Model (1)	1503.4	0.7328	0.1936	0.1880	4.367×10^{-2}	1.248×10^{-2}
Model (2)	1578.7	0.6772	0.1039	0.1964	4.490×10^{-2}	1.193×10^{-2}
Model (3)	1571.4	0.6742	0.0964	0.1836	4.468×10^{-2}	8.200×10^{-3}
Model (4)	1513.6	0.7287	0.1774	0.1893	4.418×10^{-2}	1.355×10^{-2}
Model (5)	1583.2	0.6681	0.1021	0.1968	4.506×10^{-2}	1.182×10^{-2}
Model (6)	1623.3	0.6213	0.0533	0.2013	4.522×10^{-2}	9.080×10^{-3}
Model (7)	1506.2	0.7262	0.1854	0.1892	4.388×10^{-2}	1.642×10^{-2}
Model (8)	1511.3	0.7268	0.1816	0.1894	4.391×10^{-2}	1.318×10^{-2}
Model (9)	1493.2	0.7374	0.1945	0.1874	4.383×10^{-2}	1.038×10^{-2}

The best value for each prediction accuracy measure is highlighted in bold

Table 3 Estimated model coefficients and standard errors for the logistic regression models (1), (2), and (3). Significant model coefficients at the five percent level are marked with an asterisk

	Model (1)	Model (2)	Model (3)
Intercept	7.5e+0 (3.1e+0)*	5.4e+0 (3.1e+0)	-2.2e+0 (1.1e-1)*
PLC	-6.7e-1 (2.1e-1)*		-7.1e-1 (1.9e-1)*
VAT_days	-2.2e-4 (3.6e-5)*		-1.8e-4 (3.4e-5)*
chain	-7.8e-1 (3.8e-1)*		-5.6e-1 (3.7e-1)
Coordinates	...	...	
Restaurant type	...	...	
Gprice	...	...	
Tprice	...	...	
Trating_kitchen	-3.1e-1 (3.4e-1)	-2.5e-1 (3.3e-1)	
Trating_service	-6.0e-1 (2.9e-1)*	-6.2e-1 (2.9e-1)*	
Trating_value	6.2e-1 (3.3e-1)	7.9e-1 (3.1e-1)*	
Gnr_allyears	1.1e-3 (6.4e-4)	4.4e-4 (6.8e-4)	
Grating_allyears	-2.9e-1 (3.1e-1)	-3.8e-2 (2.9e-1)	
Tnr_allyears	-6.2e-4 (1.3e-3)	-2.0e-3 (1.4e-3)	
Trating_allyears	-1.6e-1 (8.4e-2)	-1.4e-1 (8.4e-2)	
Gnr_lastyear	-2.0e-2 (4.4e-3)*	-1.6e-2 (4.4e-3)*	
Grating_lastyear	-7.8e-1 (2.8e-1)*	-7.5e-1 (2.5e-1)*	
Gsentiment_lastyear	7.7e-3 (7.5e-3)	7.8e-3 (6.9e-3)	
Tnr_lastyear	2.3e-2 (1.6e-2)	2.9e-2 (1.5e-2)	
Trating_lastyear	7.0e-2 (1.2e-1)	7.1e-2 (1.2e-1)	
Tsentiment_lastyear	-4.2e-3 (4.1e-3)	-3.5e-3 (4.1e-3)	

* $p < 0.05$

Tsentiment_lastyear), and the number of reviews and the average review rating on Google and Tripadvisor over all previous years (Gnr_allyears, Grating_allyears, Tnr_allyears, Trating_allyears) and over the last year (Gnr_lastyear, Grating_lastyear, Tnr_lastyear, Trating_lastyear). The results in Table 2 show that compared to the other models, model (1) has a high model quality, as its AIC is the second lowest, and a high prediction accuracy, as its Brier score is the lowest and its AUC and H-measure are the second highest. This set of predictor variables is therefore highly relevant for predicting restaurant bankruptcies and is based on both alternative consumer data from the restaurant platforms and official business data from the UID-register.

Model (2) includes the predictor variables from model (1) that are based on alternative consumer data and model (3) includes the predictor variables from model (1) that are based on official business data, namely the variables PLC, VAT_days, and chain, see Table 3. The AUC of 0.677 of model (2) in Table 2 confirms that it is possible to discriminate probabilistically between restaurants that will go bankrupt and those that will not based only on alternative

Table 4 Estimated model coefficients and standard errors for the logistic regression models (5), (6), and (7). Significant model coefficients at the five percent level are marked with an asterisk

	Model (5)	Model (6)	Model (7)
Intercept	4.4e+0 (3.0e+0)	4.4e+0 (3.0e+0)	1.1e+0 (4.1e+0)
PLC			-6.9e-1 (2.1e-1)*
VAT_days			-2.2e-4 (3.7e-5)*
chain			-8.0e-1 (3.9e-1)*
Coordinates	...	...	...
Restaurant type	...	...	...
Gprice	...	...	...
Tprice	...	...	...
Trating_kitchen		-3.3e-1 (3.2e-1)	-2.8e-1 (3.4e-1)
Trating_service		-6.0e-1 (2.8e-1)*	-6.1e-1 (2.9e-1)*
Trating_value		7.8e-1 (3.1e-1)*	5.9e-1 (3.3e-1)
Gnr_allyears	-1.0e-4 (6.2e-4)		2.5e-3 (1.1e-3)*
Grating_allyears	-1.7e-1 (2.7e-1)		1.8e+0 (9.2e-1)
Tnr_allyears		-3.7e-3 (1.2e-3)*	-1.2e-3 (2.4e-3)
Trating_allyears		-1.7e-1 (8.1e-2)*	-4.9e-1 (3.1e-1)
Gnr_lastyear	-1.3e-2 (4.1e-3)*		-2.0e-2 (4.4e-3)*
Grating_lastyear	-7.7e-1 (2.5e-1)*		-8.0e-1 (2.8e-1)*
Gsentiment_lastyear	7.2e-3 (6.8e-3)		8.9e-3 (7.6e-3)
Tnr_lastyear		4.0e-3 (1.6e-2)	2.4e-2 (1.6e-2)
Trating_lastyear		3.5e-2 (1.1e-1)	8.4e-2 (1.2e-1)
Tsentiment_lastyear		-3.5e-3 (4.0e-3)	-4.2e-3 (4.1e-3)
NN_distance			-9.7e-5 (7.7e-5)
NN_diff_Gnr			1.4e-3 (9.4e-4)
NN_diff_Grating			2.1e+0 (9.0e-1)*
NN_diff_Tnr			-3.5e-4 (2.3e-3)
NN_diff_Trating			-3.3e-1 (3.0e-1)
NN_similarity			-7.3e-2 (7.4e-2)

* $p < 0.05$

consumer data. Model (3) includes only three predictor variables and has a similar model quality and prediction accuracy as model (2), e.g., the AUC is identical up to the second decimal place. This highlights the importance of official business data. Specifically, the coefficients for the predictor variables PLC and VAT_days are highly significant, and if a restaurant is a PLC rather than an LLC, the predicted probability of bankruptcy is smaller. Also, the longer the restaurant's previous period of VAT liability, the smaller the predicted probability of bankruptcy. This observation is consistent with the age effect often found in bankruptcy models (Hensler et al., 1997; Altman et al., 2008). Important predictor variables based on alternative consumer data are the number of reviews (Gnr_lastyear) and the average review rating (Grating_lastyear) on Google over the past year. The corresponding model coefficients are

highly significant and, as expected, the predicted probability of bankruptcy is smaller the higher the number of reviews or the higher the average review rating over the past year. The decomposition of the predictor variables from model (1) confirms the relevance of both alternative consumer data and official business data for predicting restaurant bankruptcies, and combining these two data sources in model (1) increases the prediction accuracy. Although model (3) achieves a similar prediction accuracy as model (2) with only three traditional business predictor variables, it is important to note that alternative consumer data for restaurant bankruptcy still matters, as even small improvements in the prediction accuracy through alternative data sources can yield large economic gains. This is the case, for example, when a restaurant investor's portfolio is of high value or when the impact of a new policy affecting many restaurants is assessed.

In model (4), we extend the set of predictor variables from model (1) by adding variables measuring the number of reviews, the average review rating, and the average sentiment score over the past two and five years, see Table 10 in Appendix A.5. However, adding these predictor variables decreases model quality and prediction accuracy, and we conclude that consumer reviews from the last year are more important for bankruptcy prediction than reviews from earlier years. Further, the coefficients for the average sentiment scores over different time periods are not significant, and overall, sentiment scores have little relevance in predicting restaurant bankruptcies. There are studies on restaurant closure (Lian et al., 2017; Tao and Zhou, 2020; Li et al., 2023) that show that sentiment scores can improve prediction accuracy. Unlike our overall sentiment score, these studies assess sentiment for specific aspects, and such aspect-based sentiment analysis may be a future extension of this research.

Next, we split the predictor variables from model (2) into two subsets to compare the predictive power between the variables obtained from the Google platform and the variables obtained from the Tripadvisor platform. Specifically, in Table 4, model (5) includes only the predictor variables from Google and model (6) includes the predictor variables from Tripadvisor, the coordinates, and the variables to identify the restaurant type. Predictive bankruptcy probabilities of model (5) are more accurate than those of model (6) in terms of all prediction accuracy measures, except for the ECE, and we conclude that the variables obtained from Google have better predictive power than those obtained from Tripadvisor. Note that Google reviews over the last year are important in model (5), i.e., the coefficients for the variables *Gnr_lastyear* and *Grating_lastyear* are significant, while Tripadvisor reviews over all previous years are important in model (6), i.e., the coefficients for the variables

Tnr_allyears and *Trating_allyears* are significant. This could be due to the generally higher level of consumer activity, measured by the number of reviews, on Google than on Tripadvisor. As a result, Google reviews from the past year are rich in information, while Tripadvisor reviews contain less information that is more dispersed over time.

In model (7), we again extend the set of predictor variables from model (1) by adding variables that capture the local competition between restaurants. We observe that the coefficient for the difference in the average consumer rating on Google over all previous years between the neighborhood and the restaurant (*NN_diff_Grating*) is significant and positive. Consequently, the predicted probability of bankruptcy for a restaurant is higher if its neighboring restaurants have a higher average consumer rating on Google. We interpret this finding in the sense that stronger local restaurant competition, captured by the variable *NN_diff_Grating*, is correlated with a higher likelihood of restaurant bankruptcy. Tao and Zhou (2020) find an inverse correlation to ours for restaurant closure, i.e., the stronger the local competition, the smaller the probability of closure. Further, the coefficients for the distance (*NN_distance*) and the similarity (*NN_similarity*) of the neighboring restaurants are not significant. Compared to model (1), the additional predictor variables in model (7) lead to a decrease in model quality and prediction accuracy.

Model (8) in Table 11 in Appendix A.5 adds demographic variables to the set of predictor variables from model (1). However, none of the coefficients for the added demographic variables are significant, model quality and prediction accuracy decrease compared to model (1), and we conclude that these demographic variables are not relevant for predicting restaurant bankruptcy.

In model (9), we once more extend the set of predictor variables from model (1) by adding interaction variables between the three official business variables and the number of reviews (*Gnr_lastyear*) and the average review rating (*Grating_lastyear*) on Google over the past year. The coefficient for the interaction between the variable *chain* and *Gnr_lastyear* is significant and positive, see Table 11 in Appendix A.5. Thus, the predicted probability of bankruptcy decreases less strongly with the number of reviews on Google over the past year if the restaurant belongs to a chain. We can interpret this in the sense that consumer review data may be less informative for chain restaurants than for independent restaurants. Such an observation is also made by Luca (2016), who finds that consumer reviews affect restaurant revenues for independent restaurants but not for chain restaurants. Adding the interaction variables increases model quality, and predictive bankruptcy probabilities are more accurate than for model (1) in terms of all prediction accuracy

measures, except for the Brier score. We conclude that there are interaction effects between variables based on official business data and variables based on alternative consumer data, and that modeling these interaction effects can improve prediction accuracy.

5 Limitations

Since our data set is created exclusively from publicly available data, it is subject to certain limitations. First, the set of restaurants in the final data set is not a representative sample of the restaurant population in Switzerland, as restaurants must fulfill the restrictions described in Sect. 3.2. In particular, the restriction to PLC and LLC restaurants limits our findings. Second, adding the UIDs to the restaurants and identifying bankruptcies is a non-trivial task because every company operating in Switzerland is assigned a UID and a company can have more than one local business. Consequently, for restaurant chains, the potential bankruptcy relates to the entire company and not to the individual restaurant. Because we do not include restaurants that are part of a cooperative, the number of chain restaurants in our data set is limited. In particular, we find that the UID of 455 restaurants appears more than once in our data set and these restaurants belong to 153 different businesses. A third limitation is that our data set does not include the financial ratios of the restaurants, which are traditionally used in bankruptcy models. We therefore cannot conclusively show that the use of alternative consumer data in addition to traditional business data leads to more accurate restaurant bankruptcy predictions, although our results provide strong evidence in this direction.

6 Conclusion

We create a unique data set for predicting bankruptcies of individual restaurants in Switzerland by collecting alternative consumer data from restaurant platforms and official

business data from Swiss government websites. In evaluating the accuracy of predictive bankruptcy probabilities, we find that alternative consumer data allows for probabilistic discrimination between restaurants that will go bankrupt and those that will not. The highest prediction accuracy, with an AUC of 0.737, is achieved when alternative consumer data and official business data are combined. In our models, the number of consumer reviews and the average review rating on Google over the last year are important predictor variables, and higher values of these two variables reduce the predicted probability of bankruptcy. We also find that consumer reviews from the last year are more important for restaurant bankruptcy prediction than reviews from earlier years, and consumer data from Google is more predictive than consumer data from Tripadvisor. Restaurant competition can effectively be measured by the difference in the average consumer rating between the neighborhood and the restaurant, and interaction effects between predictor variables based on official business data and predictor variables based on alternative consumer data exist and can improve the prediction accuracy of bankruptcy models.

Appendix

Tuning parameters

Table 5 Candidate tuning parameters for tree-boosting

Tuning parameter	Candidate values
Number of trees	{1,2,...,1000}
Learning rate	{1,0.1,0.01}
Maximal tree depth	{2,3,5,10}
Minimal number of samples per leaf	{10,100,1000}
L2 regularization	{0,1,10}

Examples of sentiment scores for review texts

Table 6 Sentiment scores for a positive, a neutral, and a negative review text

Sentiment score	Review text
97.3	"Mein Lieblingsrestaurant in Zürich. Die Knödel sind nicht von dieser Welt."
49.6	"Ein Restaurant direkt in der Nähe vom Hauptbahnhof und an der Sihl, so bleibt die Kundschaft nicht aus. Was jedoch auch dazu beiträgt, dass man mit langen Wartezeiten an der Schlange stehen muss, um ein Getränk zu bestellen. Bei gutem Wetter kann man draussen sitzen und sich dort kulinarisch verwöhnen lassen."
2.1	"Mitleidiger Service, Preis-Leistungsverhältnis zu teuer für wenig Menge! Nicht empfehlenswert für ein indisches Restaurant!"

Predictor variables

Table 7 Description of predictor variables

Business variables	
PLC	Indicates whether the restaurant business is a public limited company.
VAT_days	Number of days for which the restaurant business is liable to VAT.
chain	Indicates whether at least one other restaurant in the data set has the same UID.
Coordinates	
longitude	Longitude coordinate of the restaurant
latitude	Latitude coordinate of the restaurant
Restaurant type	
swiss_kitchen	Indicates whether the restaurant serves Swiss food.
italian_kitchen	Indicates whether the restaurant serves Italian food.
thai_kitchen	Indicates whether the restaurant serves Thai food.
chinese_kitchen	Indicates whether the restaurant serves Chinese food.
indian_kitchen	Indicates whether the restaurant serves Indian food.
asian_kitchen	Indicates whether the restaurant serves Asian food.
japanese_kitchen	Indicates whether the restaurant serves Japanese food.
pizza	Indicates whether the restaurant serves pizza.
burger	Indicates whether the restaurant serves burgers.
sushi	Indicates whether the restaurant serves sushi.
steak	Indicates whether the restaurant serves steak.
cafe	Indicates whether the restaurant is a café.
bar	Indicates whether the restaurant is a bar.
bistro	Indicates whether the restaurant is a bistro.
take_away	Indicates whether the restaurant is a take-away.
Price categories	
Gprice	Price category on Google with levels 0,1,2,3 and 4. Higher levels generally mean higher prices, except for level 0, which is used for missing values.
Tprice	Price category on Tripadvisor with levels 0,1,2 and 3. Higher levels generally mean higher prices, except for level 0, which is used for missing values.
Variables based on reviews from all previous years	
Trating_kitchen	Rating of the quality of the kitchen on Tripadvisor.
Trating_service	Rating of the quality of service on Tripadvisor.
Trating_value	Rating of the value for money on Tripadvisor.
Gnr_allyears	Number of reviews on Google in all previous years
Grating_allyears	Average rating on Google over all previous years

Table 7 (continued)

Tnr_allyears	Number of reviews on Tripadvisor in all previous years
Trating_allyears	Average rating on Tripadvisor over all previous years
Variables based on reviews from the past year	
Gnr_lastyear	Number of reviews on Google in the past year
Grating_lastyear	Average rating on Google over the past year
Gsentiment_lastyear	Average sentiment score (higher = better) on Google over the past year
Tnr_lastyear	Number of reviews on Tripadvisor in the past year
Trating_lastyear	Average rating on Tripadvisor over the past year
Tsentiment_lastyear	Average sentiment score (higher = better) on Tripadvisor over the past year
Variables based on reviews from the past two years	
Gnr_last2years	Number of reviews on Google in the past two years
Grating_last2years	Average rating on Google over the past two years
Gsentiment_last2years	Average sentiment score (higher = better) on Google over the past two years
Tnr_last2years	Number of reviews on Tripadvisor in the past two years
Trating_last2years	Average rating on Tripadvisor over the past two years
Tsentiment_last2years	Average sentiment score (higher = better) on Tripadvisor over the past two years
Variables based on reviews from the past five years	
Gnr_last5years	Number of reviews on Google in the past five years
Grating_last5years	Average rating on Google over the past five years
Gsentiment_last5years	Average sentiment score (higher = better) on Google over the past five years
Tnr_last5years	Number of reviews on Tripadvisor in the past five years
Trating_last5years	Average rating on Tripadvisor over the past five years
Tsentiment_last5years	Average sentiment score (higher = better) on Tripadvisor over the past five years
Neighborhood variables	
NN_distance	Average distance in meters to the twenty closest restaurants
NN_diff_Gnr	Average number of reviews on Google for the twenty closest restaurants, minus the restaurant's number of reviews on Google (Gnr_allyears)
NN_diff_Grating	Average rating on Google over all previous years for the twenty closest restaurants, minus the restaurant's average rating on Google (Grating_allyears)
NN_diff_Tnr	Average number of reviews on Tripadvisor for the twenty closest restaurants, minus the restaurant's number of reviews on Tripadvisor (Tnr_allyears)
NN_diff_Trating	Average rating on Tripadvisor over all previous years for the twenty closest restaurants, minus the restaurant's average rating on Tripadvisor (Trating_allyears)
NN_similarity	Number of restaurants out of the twenty closest restaurants with the same restaurant type
Demographic variables	
population_size	Number of residents in the municipality
employed_first	Number of first sector employees in the municipality
employed_second	Number of second sector employees in the municipality
employed_third	Number of third sector employees in the municipality
region_income	Median gross monthly wage in the greater region
canton_tax	Cantonal tax on profits in percent
Interaction variables	
PLC_Grating	Interaction between variables PLC and Grating_lastyear
PLC_Gnr	Interaction between variables PLC and Gnr_lastyear
VAT_days_Grating	Interaction between variables VAT_days and Grating_lastyear
VAT_days_Gnr	Interaction between variables VAT_days and Gnr_lastyear
chain_Grating	Interaction between variables chain and Grating_lastyear
chain_Gnr	Interaction between variables chain and Gnr_lastyear

Summary statistics of predictor variables

Table 8 Summary statistics for numeric predictor variables for all 4'275 observations

	Min.	Q.25%	Median	Mean	Q.75%	Max.
VAT_days	12.00	1520.00	3165.00	3998.57	6035.00	19783.00
longitude	2494279.01	2601511.40	2665954.04	2645666.32	2688764.66	2825246.59
latitude	1076710.51	1163242.44	1224314.06	1205589.52	1249110.07	1292191.05
Trating_kitchen	1.50	4.00	4.20	4.20	4.50	5.00
Trating_service	2.00	4.00	4.14	4.14	4.50	5.00
Trating_value	1.50	3.93	3.93	3.93	4.00	5.00
Gnr_allyears	1.00	100.00	191.00	287.99	359.00	5081.00
Grating_allyears	1.00	4.20	4.40	4.39	4.60	5.00
Tnr_allyears	0.00	10.00	34.00	82.91	98.00	3390.00
Trating_allyears	1.00	4.00	4.00	4.00	4.50	5.00
Gnr_lastyear	0.00	17.00	33.00	47.89	59.00	1752.00
Grating_lastyear	1.00	4.26	4.50	4.43	4.71	5.00
Gsentiment_lastyear	0.02	77.59	87.10	83.87	94.64	99.98
Tnr_lastyear	0.00	0.00	2.00	4.10	5.00	96.00
Trating_lastyear	1.00	3.83	3.98	3.98	4.69	5.00
Tsentiment_lastyear	0.01	75.71	75.71	75.85	99.80	99.98
Gnr_last2years	0.00	34.00	61.00	86.38	108.00	1816.00
Grating_last2years	1.00	4.28	4.50	4.44	4.68	5.00
Gsentiment_last2years	0.14	78.75	87.26	84.32	93.45	99.98
Tnr_last2years	0.00	1.00	5.00	9.19	11.00	193.00
Trating_last2years	1.00	3.71	4.03	4.04	4.67	5.00
Tsentiment_last2years	0.01	75.27	77.17	77.35	99.73	99.98
Gnr_last5years	0.00	95.00	180.00	271.43	339.00	4595.00
Grating_last5years	1.00	4.24	4.43	4.40	4.59	5.00
Gsentiment_last5years	0.00	80.31	87.05	84.92	92.00	99.98
Tnr_last5years	0.00	7.00	22.00	44.35	54.00	1855.00
Trating_last5years	1.00	3.82	4.18	4.12	4.55	5.00
Tsentiment_last5years	0.02	74.96	81.73	80.54	96.91	99.98
NN_distance	34.84	123.95	353.44	917.36	1302.59	10046.09
NN_diff_Gnr	-4616.75	-135.05	15.90	-52.23	113.75	953.90
NN_diff_Grating	-0.97	-0.22	-0.04	-0.02	0.14	3.25
NN_diff_Tnr	-3250.50	-42.65	7.20	-21.95	34.55	465.55
NN_diff_Trating	-2.08	-0.67	-0.27	-0.14	0.17	3.50
NN_similarity	0.00	0.00	0.00	0.74	1.00	11.00
population_size	51.00	4057.00	18139.00	96032.43	139408.00	420217.00
employed_first	0.00	38.00	96.00	120.84	177.00	751.00
employed_second	0.00	466.00	2317.00	8177.02	11861.00	34946.00
employed_third	12.00	1149.50	10434.00	95892.34	150459.00	462410.00
region_income	5590.00	6779.00	6849.00	6807.78	6969.00	7229.00
canton_tax	11.85	14.00	17.42	16.88	19.65	21.04
PLC_Grating	0.00	0.00	0.00	1.57	4.26	5.00
PLC_Gnr	0.00	0.00	0.00	21.15	25.00	892.00
VAT_days_Grating	57.06	6821.48	13794.15	17551.37	26105.64	95610.00
VAT_days_Gnr	0.00	32823.50	92138.00	204734.26	230191.50	8421864.00
chain_Grating	0.00	0.00	0.00	0.46	0.00	5.00
chain_Gnr	0.00	0.00	0.00	7.27	0.00	424.00

Table 9 Summary statistics for categorical predictor variables for all 4'275 observations

	Level	Count
PLC	0	2733
	1	1542
chain	0	3820
	1	455
Gprice	0	1553
	1	214
	2	2245
	3	237
	4	26
Tprice	0	670
	1	287
	2	3031
	3	287
swiss_kitchen	0	3791
	1	484
italian_kitchen	0	3912
	1	363
thai_kitchen	0	4190
	1	85
chinese_kitchen	0	4203
	1	72
indian_kitchen	0	4210
	1	65
asian_kitchen	0	4230
	1	45
japanese_kitchen	0	4239
	1	36
pizza	0	4010
	1	265
burger	0	4236
	1	39
sushi	0	4235
	1	40
steak	0	4241
	1	34
cafe	0	4166
	1	109
bar	0	4028
	1	247
bistro	0	4240
	1	35
take_away	0	4236
	1	39

Coefficient tables

Table 10 Estimated model coefficients and standard errors for the logistic regression models (1) to (4). Significant coefficients at the five percent level are marked with an asterisk

	Model (1)	Model (2)	Model (3)	Model (4)
Intercept	7.5e+0 (3.1e+0)*	5.4e+0 (3.1e+0)	-2.2e+0 (1.1e-1)*	8.7e+0 (3.2e+0)*
PLC	-6.7e-1 (2.1e-1)*		-7.1e-1 (1.9e-1)*	-6.7e-1 (2.1e-1)*
VAT_days	-2.2e-4 (3.6e-5)*		-1.8e-4 (3.4e-5)*	-2.3e-4 (3.7e-5)*
chain	-7.8e-1 (3.8e-1)*		-5.6e-1 (3.7e-1)	-7.6e-1 (3.9e-1)*
longitude	-2.9e-6 (1.3e-6)*	-3.1e-6 (1.3e-6)*		-3.0e-6 (1.3e-6)*
latitude	3.7e-6 (1.7e-6)*	3.5e-6 (1.7e-6)*		3.6e-6 (1.7e-6)*
swiss_kitchen	5.9e-1 (2.4e-1)*	5.5e-1 (2.4e-1)*		5.8e-1 (2.4e-1)*
italian_kitchen	8.7e-1 (2.4e-1)*	8.6e-1 (2.4e-1)*		8.9e-1 (2.4e-1)*
thai_kitchen	-7.7e-1 (7.3e-1)	-6.1e-1 (7.3e-1)		-8.2e-1 (7.4e-1)
chinese_kitchen	-3.7e-1 (7.4e-1)	-5.5e-1 (7.4e-1)		-3.9e-1 (7.5e-1)
indian_kitchen	-1.3e+0 (1.0e+0)	-1.0e+0 (1.0e+0)		-1.3e+0 (1.0e+0)
asian_kitchen	8.6e-1 (5.6e-1)	6.1e-1 (5.6e-1)		9.2e-1 (5.7e-1)
japanese_kitchen	-3.5e-2 (1.0e+0)	-2.0e-1 (1.0e+0)		-2.1e-2 (1.0e+0)
pizza	2.9e-1 (2.9e-1)	2.9e-1 (2.9e-1)		3.5e-1 (3.0e-1)
burger	-4.1e-1 (1.1e+0)	-2.1e-1 (1.0e+0)		-3.1e-1 (1.1e+0)
sushi	-8.5e-1 (1.0e+0)	-7.1e-1 (1.0e+0)		-7.4e-1 (1.0e+0)
steak	8.9e-1 (7.6e-1)	9.1e-1 (7.6e-1)		9.7e-1 (7.7e-1)
cafe	-1.5e-1 (5.3e-1)	-2.5e-1 (5.3e-1)		-9.2e-2 (5.4e-1)
bar	1.3e-1 (3.3e-1)	8.3e-2 (3.2e-1)		1.3e-1 (3.3e-1)
bistro	4.8e-1 (7.5e-1)	3.9e-1 (7.5e-1)		5.8e-1 (7.6e-1)
take_away	-1.3e+1 (3.6e+2)	-1.3e+1 (3.7e+2)		-1.4e+1 (3.6e+2)
Gprice_1	2.7e-1 (3.3e-1)	3.1e-1 (3.2e-1)		2.3e-1 (3.3e-1)
Gprice_2	-1.9e-1 (1.8e-1)	-2.1e-1 (1.8e-1)		-2.3e-1 (1.9e-1)
Gprice_3	-1.5e+0 (7.6e-1)*	-1.5e+0 (7.5e-1)*		-1.6e+0 (7.6e-1)*
Gprice_4	1.1e+0 (8.6e-1)	8.8e-1 (8.5e-1)		1.2e+0 (8.7e-1)
Tprice_1	-1.1e+0 (4.7e-1)*	-1.2e+0 (4.6e-1)*		-1.0e+0 (4.7e-1)*
Tprice_2	8.8e-2 (2.1e-1)	-7.2e-3 (2.1e-1)		1.2e-1 (2.1e-1)
Tprice_3	2.7e-1 (4.2e-1)	1.1e-2 (4.2e-1)		2.9e-1 (4.3e-1)
Trating_kitchen	-3.1e-1 (3.4e-1)	-2.5e-1 (3.3e-1)		-2.4e-1 (3.4e-1)
Trating_service	-6.0e-1 (2.9e-1)*	-6.2e-1 (2.9e-1)*		-5.8e-1 (3.0e-1)
Trating_value	6.2e-1 (3.3e-1)	7.9e-1 (3.1e-1)*		5.8e-1 (3.3e-1)
Gnr_allyears	1.1e-3 (6.4e-4)	4.4e-4 (6.8e-4)		7.4e-4 (6.2e-3)
Grating_allyears	-2.9e-1 (3.1e-1)	-3.8e-2 (2.9e-1)		2.8e-1 (1.3e+0)
Tnr_allyears	-6.2e-4 (1.3e-3)	-2.0e-3 (1.4e-3)		2.1e-3 (2.6e-3)
Trating_allyears	-1.6e-1 (8.4e-2)	-1.4e-1 (8.4e-2)		-1.7e-1 (9.2e-2)
Gnr_lastyear	-2.0e-2 (4.4e-3)*	-1.6e-2 (4.4e-3)*		-2.3e-2 (1.0e-2)*
Grating_lastyear	-7.8e-1 (2.8e-1)*	-7.5e-1 (2.5e-1)*		-3.4e-1 (3.3e-1)
Gsentiment_lastyear	7.7e-3 (7.5e-3)	7.8e-3 (6.9e-3)		-9.4e-3 (9.6e-3)
Tnr_lastyear	2.3e-2 (1.6e-2)	2.9e-2 (1.5e-2)		4.1e-2 (3.3e-2)
Trating_lastyear	7.0e-2 (1.2e-1)	7.1e-2 (1.2e-1)		3.0e-2 (1.5e-1)
Tsentiment_lastyear	-4.2e-3 (4.1e-3)	-3.5e-3 (4.1e-3)		6.0e-4 (5.2e-3)
Gnr_last2years				2.5e-3 (6.4e-3)
Grating_last2years				-8.7e-1 (4.7e-1)
Gsentiment_last2years				2.4e-2 (1.5e-2)
Tnr_last2years				-3.1e-3 (2.1e-2)
Trating_last2years				4.0e-2 (1.8e-1)

Table 10 (continued)

	Model (1)	Model (2)	Model (3)	Model (4)
Tsentiment_last2years				-7.9e-3 (5.6e-3)
Gnr_last5years				2.9e-4 (6.9e-3)
Grating_last5years				-9.6e-1 (1.3e+0)
Gsentiment_last5years				2.2e-2 (1.6e-2)
Tnr_last5years				-6.9e-3 (6.4e-3)
Trating_last5years				6.3e-2 (2.1e-1)
Tsentiment_last5years				1.6e-4 (5.6e-3)

* $p < 0.05$ **Table 11** Estimated model coefficients and standard errors for the logistic regression models (5) to (9). Significant coefficients at the five percent level are marked with an asterisk

	Model (5)	Model (6)	Model (7)	Model (8)	Model (9)
Intercept	4.4e+0 (3.0e+0)	4.4e+0 (3.0e+0)	1.1e+0 (4.1e+0)	1.1e+1 (4.5e+0)*	9.3e+0 (3.3e+0)*
PLC			-6.9e-1 (2.1e-1)*	-6.9e-1 (2.1e-1)*	-2.2e+0 (1.4e+0)
VAT_days			-2.2e-4 (3.7e-5)*	-2.2e-4 (3.7e-5)*	-7.0e-4 (3.0e-4)*
chain			-8.0e-1 (3.9e-1)*	-8.1e-1 (3.9e-1)*	1.0e+0 (2.5e+0)
longitude	-2.9e-6 (1.2e-6)*	-3.1e-6 (1.3e-6)*	-3.0e-6 (1.3e-6)*	-3.8e-6 (1.8e-6)*	-2.9e-6 (1.3e-6)*
latitude	3.5e-6 (1.6e-6)*	2.2e-6 (1.7e-6)	2.6e-6 (1.8e-6)	5.3e-6 (2.7e-6)	3.6e-6 (1.7e-6)*
swiss_kitchen	5.4e-1 (2.4e-1)*	3.8e-1 (2.3e-1)	8.1e-1 (3.2e-1)*	6.3e-1 (2.5e-1)*	6.0e-1 (2.5e-1)*
italian_kitchen	8.3e-1 (2.3e-1)*	7.6e-1 (2.3e-1)*	9.3e-1 (2.7e-1)*	8.8e-1 (2.4e-1)*	8.7e-1 (2.4e-1)*
thai_kitchen	-7.4e-1 (7.3e-1)	-5.5e-1 (7.3e-1)	-7.2e-1 (7.4e-1)	-8.0e-1 (7.3e-1)	-8.3e-1 (7.4e-1)
chinese_kitchen	-5.7e-1 (7.3e-1)	-4.5e-1 (7.3e-1)	-4.5e-1 (7.4e-1)	-3.3e-1 (7.4e-1)	-4.0e-1 (7.4e-1)
indian_kitchen	-1.1e+0 (1.0e+0)	-9.2e-1 (1.0e+0)	-1.3e+0 (1.0e+0)	-1.3e+0 (1.0e+0)	-1.3e+0 (1.0e+0)
asian_kitchen	6.0e-1 (5.5e-1)	7.4e-1 (5.4e-1)	8.9e-1 (5.7e-1)	9.3e-1 (5.6e-1)	9.1e-1 (5.7e-1)
japanese_kitchen	-2.1e-1 (1.0e+0)	-4.6e-1 (1.0e+0)	-2.8e-2 (1.0e+0)	-7.6e-2 (1.0e+0)	-1.5e-2 (1.0e+0)
pizza	2.7e-1 (2.9e-1)	3.1e-1 (2.8e-1)	3.8e-1 (3.1e-1)	3.2e-1 (3.0e-1)	2.8e-1 (3.0e-1)
burger	-1.8e-1 (1.0e+0)	-5.4e-1 (1.0e+0)	-4.5e-1 (1.1e+0)	-4.9e-1 (1.1e+0)	-3.9e-1 (1.1e+0)
sushi	-6.0e-1 (1.0e+0)	-8.4e-1 (1.0e+0)	-9.5e-1 (1.0e+0)	-8.5e-1 (1.0e+0)	-8.1e-1 (1.1e+0)
steak	8.7e-1 (7.5e-1)	5.8e-1 (7.5e-1)	8.6e-1 (7.7e-1)	9.4e-1 (7.7e-1)	1.0e+0 (7.7e-1)
cafe	-2.7e-1 (5.3e-1)	-2.6e-1 (5.2e-1)	-1.1e-1 (5.4e-1)	-1.8e-1 (5.3e-1)	-7.8e-2 (5.3e-1)
bar	6.8e-2 (3.2e-1)	3.9e-2 (3.2e-1)	1.8e-1 (3.6e-1)	8.2e-2 (3.3e-1)	1.8e-1 (3.3e-1)
bistro	2.7e-1 (7.4e-1)	4.5e-1 (7.4e-1)	4.5e-1 (7.6e-1)	4.8e-1 (7.6e-1)	5.2e-1 (7.6e-1)
take_away	-1.4e+1 (3.8e+2)	-1.3e+1 (3.7e+2)	-1.3e+1 (3.6e+2)	-1.4e+1 (3.6e+2)	-1.3e+1 (3.6e+2)
Gprice_1	1.3e-1 (3.1e-1)		3.1e-1 (3.3e-1)	2.5e-1 (3.3e-1)	2.7e-1 (3.3e-1)
Gprice_2	-1.8e-1 (1.7e-1)		-1.9e-1 (1.8e-1)	-1.9e-1 (1.8e-1)	-1.6e-1 (1.8e-1)
Gprice_3	-1.8e+0 (7.3e-1)*		-1.6e+0 (7.6e-1)*	-1.5e+0 (7.6e-1)*	-1.5e+0 (7.6e-1)*
Gprice_4	5.6e-1 (7.6e-1)		1.1e+0 (8.7e-1)	1.1e+0 (8.6e-1)	1.2e+0 (8.6e-1)
Tprice_1		-1.2e+0 (4.5e-1)*	-1.2e+0 (4.7e-1)*	-1.2e+0 (4.7e-1)*	-1.1e+0 (4.7e-1)*
Tprice_2		-1.8e-1 (2.0e-1)	1.1e-1 (2.1e-1)	8.4e-2 (2.1e-1)	1.5e-1 (2.1e-1)
Tprice_3		-3.8e-2 (3.9e-1)	3.1e-1 (4.3e-1)	2.8e-1 (4.2e-1)	3.4e-1 (4.3e-1)
Trating_kitchen		-3.3e-1 (3.2e-1)	-2.8e-1 (3.4e-1)	-3.1e-1 (3.4e-1)	-3.4e-1 (3.4e-1)
Trating_service		-6.0e-1 (2.8e-1)*	-6.1e-1 (2.9e-1)*	-6.1e-1 (2.9e-1)*	-6.1e-1 (2.9e-1)*
Trating_value		7.8e-1 (3.1e-1)*	5.9e-1 (3.3e-1)	6.4e-1 (3.3e-1)	6.4e-1 (3.3e-1)
Gnr_allyears	-1.0e-4 (6.2e-4)		2.5e-3 (1.1e-3)*	1.1e-3 (6.4e-4)	9.5e-4 (6.7e-4)

Table 11 (continued)

	Model (5)	Model (6)	Model (7)	Model (8)	Model (9)
Grating_allyears	-1.7e-1 (2.7e-1)		1.8e+0 (9.2e-1)	-2.5e-1 (3.1e-1)	-2.6e-1 (3.3e-1)
Tnr_allyears		-3.7e-3 (1.2e-3)*	-1.2e-3 (2.4e-3)	-7.2e-4 (1.3e-3)	-7.6e-4 (1.3e-3)
Trating_allyears		-1.7e-1 (8.1e-2)*	-4.9e-1 (3.1e-1)	-1.7e-1 (8.5e-2)*	-1.5e-1 (8.6e-2)
Gnr_lastyear	-1.3e-2 (4.1e-3)*		-2.0e-2 (4.4e-3)*	-1.9e-2 (4.4e-3)*	-2.4e-2 (5.2e-3)*
Grating_lastyear	-7.7e-1 (2.5e-1)*		-8.0e-1 (2.8e-1)*	-8.0e-1 (2.8e-1)*	-1.2e+0 (3.7e-1)*
Gsentiment_lastyear	7.2e-3 (6.8e-3)		8.9e-3 (7.6e-3)	7.4e-3 (7.6e-3)	8.9e-3 (7.8e-3)
Tnr_lastyear		4.0e-3 (1.6e-2)	2.4e-2 (1.6e-2)	2.3e-2 (1.6e-2)	2.4e-2 (1.5e-2)
Trating_lastyear		3.5e-2 (1.1e-1)	8.4e-2 (1.2e-1)	8.2e-2 (1.2e-1)	8.3e-2 (1.2e-1)
Tsentiment_lastyear		-3.5e-3 (4.0e-3)	-4.2e-3 (4.1e-3)	-4.5e-3 (4.1e-3)	-4.7e-3 (4.2e-3)
NN_distance			-9.7e-5 (7.7e-5)		
NN_diff_Gnr			1.4e-3 (9.4e-4)		
NN_diff_Grating			2.1e+0 (9.0e-1)*		
NN_diff_Tnr			-3.5e-4 (2.3e-3)		
NN_diff_Trating			-3.3e-1 (3.0e-1)		
NN_similarity			-7.3e-2 (7.4e-2)		
population_size				4.8e-6 (5.9e-6)	
employed_first				4.9e-4 (8.0e-4)	
employed_second				3.7e-6 (1.6e-5)	
employed_third				-4.0e-6 (5.0e-6)	
region_income				-4.0e-4 (3.4e-4)	
canton_tax				-1.3e-3 (2.9e-2)	
PLC_Grating					2.8e-1 (3.3e-1)
PLC_Gnr					1.1e-2 (5.8e-3)
VAT_days_Grating					1.2e-4 (6.8e-5)
VAT_days_Gnr					-9.7e-7 (1.2e-6)
chain_Grating					-7.0e-1 (6.2e-1)
chain_Gnr					2.0e-2 (6.1e-3)*

* $p < 0.05$ **Abbreviations**

FSO	Swiss Federal Statistical Office
UID	Enterprise identification number
VAT	Value-added tax
SOGC	Swiss Official Gazette of Commerce
PLC	Public limited company
LLC	Limited liability company
AIC	Akaike Information Criterion
AUC	Area under the receiver operating characteristic curve
ECE	Expected calibration error
GP	Gaussian process

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Data availability

The data set generated and analyzed in the study is not publicly available due to the terms and conditions of the online restaurant platforms.

Declarations**Conflict of interest**

The author declares that he has no conflict of interest.

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